



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

Assignment 4 – FALL 2025

Course Title:	Microprocessor Systems and Interfacing				Course Code:	CPE 342	Credit Hours:	4(3,1)
Course Instructor:	Dr. Abbas Javed, Mr. Moazzam Ali Sahi				Program Name:	BCE		
Semester:	5 th	Batch:	FA23	Section:	A & B	Date Given:	December 12, 2025	
Submission Date:	December 19, 2025				Maximum Marks:	30		
Name:					Registration Number:			
<u>Important Instructions / Guidelines:</u>								
• Do your own work, PLAGARISM will be graded as ZERO. • No late submission.								

Question 1: [CLO2-PLO3-C5] [3+10+2 Marks]

Design an 8086-microprocessor based temperature controller for maintaining indoor thermal comfort. Temperature controller should control the HVAC to maintain the cooling setpoint entered by the occupant. Temperature controller should turn OFF the HVAC when indoor room temperature is equal to or less than cooling setpoint and turn ON the HVAC when the room temperature is greater than cooling setpoint. As a design engineer, you are required to implement the following tasks:

1. Interface a LM 35 temperature sensor using Analog to Digital Converter (ADC) with 8086 microprocessor. The LM35 temperature sensor provides an analog voltage output proportional to the temperature in degrees Celsius (10 mV per °C). Assume reference voltage of 5.0 V and starting address of 700H. The output of ADC (Digital Value) can be converted to temperature by the following Equation 1:
$$Temperature = \left(\frac{Digital\ Value \times 500}{255} \right) \quad (1)$$
2. Integrate a suitable peripheral for allowing the occupant to enter the required cooling setpoint.
3. Control the HVAC using digital switch connected to the digital output of the microprocessor.

Provide:

- a) Completely labelled schematic diagram.
- b) Assembly code for designing the temperature controller.